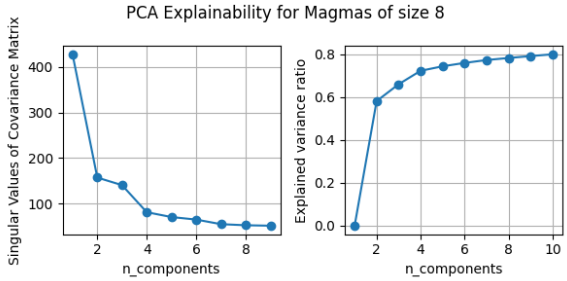

Figure 4: PCA Explainability statistics for a random sample of $n = 20\,000$ magmas of size 8. The values of the first ten singular values (left) and the explained variance ratio (right) for the first ten principal components.

choice of picking the three first dimensions of the PCA is justified by our visualization purposes as well as by the fact that a majority of variability in the data is captured by the first three principal components, as shown in Fig. 4.

6.2 The latent space is good at detecting signatures of equational theories

Exploring the latent space and looking at each position of the 4964 equational theories, we observe that the latent space is surprisingly good at identifying and grouping together the equational theories $\mathsf{Eqn}[k]$ with the same signature, see Fig. 5.

Here, by signature of an equational theory, we mean the ordered pair (a, b) describing the numbers a and b of diamonds on the left-hand side and right-hand side of the equation, respectively. Typically, the Tarski's law $\mathsf{Eqn}[542]$

$$x = (z \diamond (x \diamond (y \diamond z)))$$

has signature $(0, 3)$. We indicate how the 4694 equational theories are partitioned according to their signature, and how they are colored in the latent space depicted in Fig. 5.

2842	theories have signature	$(0,4)$	[red]
1015	theories have signature	$(1,3)$	[blue]
427	theories have signature	$(2,2)$	[green]
260	theories have signature	$(0,3)$	[purple]
104	theories have signature	$(1,2)$	[cyan]
30	theories have signature	$(0,2)$	[pink]
9	theories have signature	$(1,1)$	[yellow]
5	theories have signature	$(0,1)$	[pink]
2	theories have signature	$(0,0)$	[pink]

6.3 Expectation and variance

The *expectation* and *variance* of an equational theory $\mathsf{Eqn}[k]$ are defined as the expectation and variance of the associated sequence of Stone pairings:

$$E[\,\mathsf{Eqn}[k]\,] = \tfrac{1}{n}\textstyle\sum_{i=1}^{n} p_i$$
$$V[\,\mathsf{Eqn}[k]\,] = \tfrac{1}{n}\textstyle\sum_{i=1}^{n} \Big(p_i - E[\,\mathsf{Eqn}[k]\,]\Big)^2$$

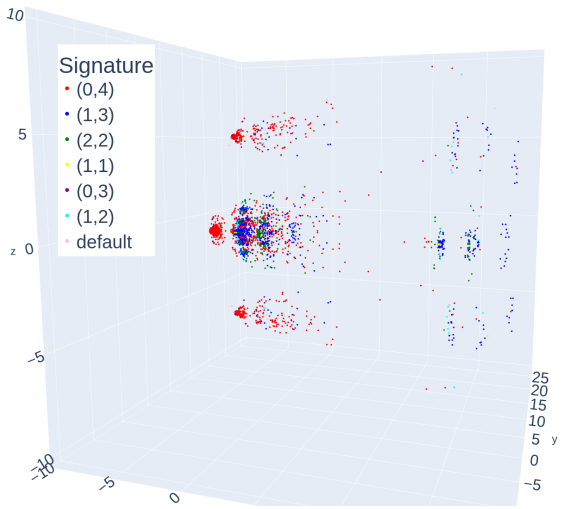

Figure 5: Latent space of equational theories where every theory is colored according to its signature (a, b), where a and b are the numbers of diamond operations on the lefthand and righthand side of the theory.

where $p_i = \langle\, \mathsf{Eqn}[k] \mid A_i \,\rangle$. Note that the expectation and variance do not depend on the sampling order of the finite magmas $A_1, \ldots, A_n$ and can be thus computed from the Stone spectrum of $\mathsf{Eqn}[k]$.

6.4 The X-axis is strongly correlated to expectation

The first principal component X of a theory $\mathsf{Eqn}[k]$ is strongly correlated to its expectation $E[\,\mathsf{Eqn}[k]\,]$, as shown by the linear regression graph, see Fig. 8 in Appendix. As a consequence, the theory with maximum X-value is $\mathsf{Eqn}[0]$

$$x = x$$

with expectation exactly 1, followed by $\mathsf{Eqn}[3277]$ and its conjugate $\mathsf{Eqn}[4067]$

$$x \diamond x = ((x \diamond x) \diamond y) \diamond y$$
$$x \diamond x = y \diamond (y \diamond (x \diamond x))$$

whose expectation in the sample of finite magmas of size 8 we took is around .46. On the other side of the latent space, one of the equational theories with smallest X-value is $\mathsf{Eqn}[1]$

$$x = y$$

with expectation exactly $\frac{1}{8} = .125$ in any finite magma of size $N = 8$ used in the sample.

6.5 The Y-axis is correlated to variance

The principal component Y of an equational theory $\mathsf{Eqn}[k]$ is strongly correlated to its variance $V[\,\mathsf{Eqn}[k]\,]$, as shown by the linear regression, see Fig. 9 in Appendix. The equational theory with minimum variance and Y-value is $\mathsf{Eqn}[4275]$ and conjugate $\mathsf{Eqn}[4590]$: